

		This expression is equal to the sum of clockwise moments = $(20 + 25)r$.
6	A	Net work done = change in KE First hit: $-F_f (1/4 d) = 0 - \frac{1}{2} m u^2 \quad \text{--- (1)}$ Second hit: $-F_f (d) = 0 - \frac{1}{2} m u_1^2 \quad \text{--- (2)}$ Thus, $(2) / (1) \Rightarrow 4 = u_1^2 / u^2$ $\Rightarrow u_1 = 2 u$
7	D	By C O M E